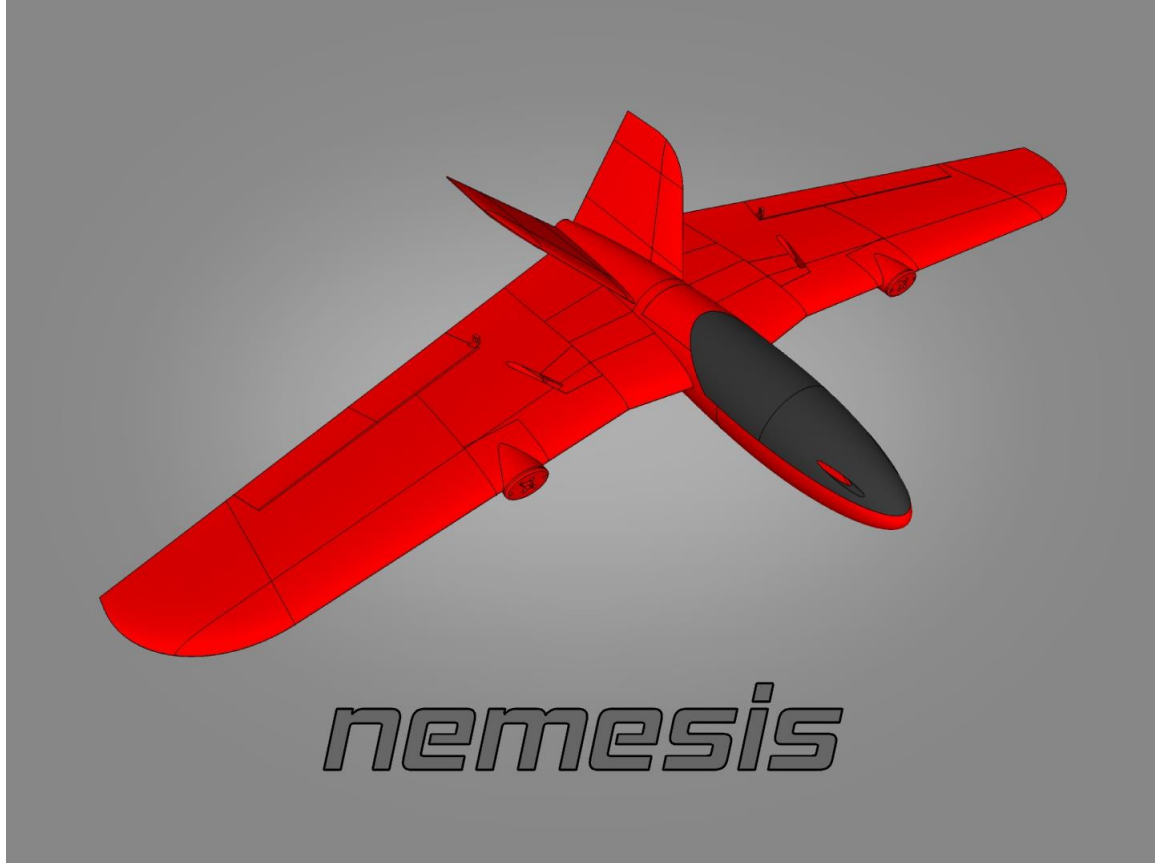


Nemesis Instruction Manual

Last updated: September 27th, 2025



The Nemesis is a 3D printed RC model created by [Stunt Double](#). This model is a 1.2m wingspan forward-swept flying wing with twin tractor motors. It's a fast highly stable cruiser designed for FPV or LOS flying.

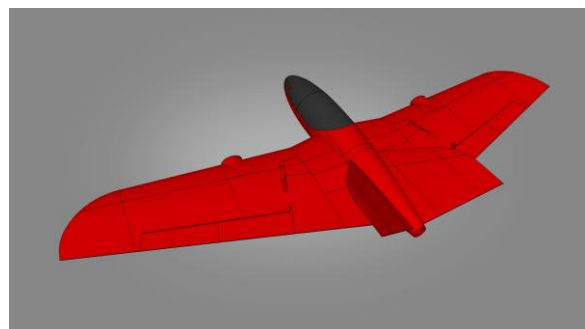
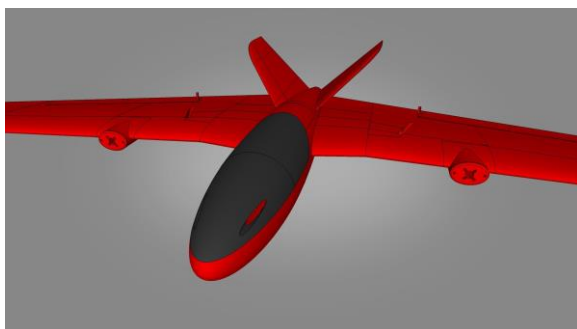
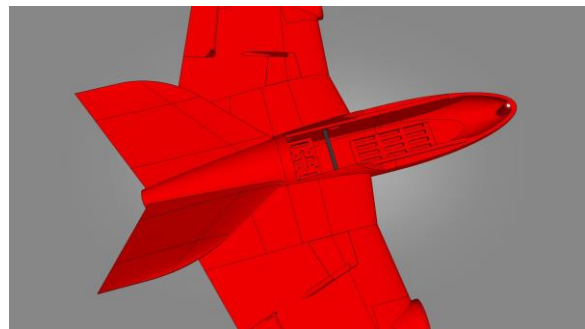
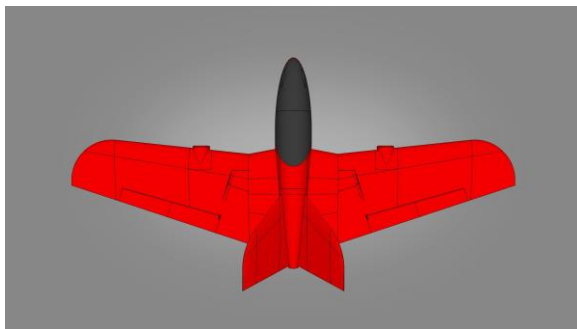
This document provides all the necessary information to build the Nemesis. It includes information about the equipment required, printing settings, slicer profiles, assembly instructions, setup details & more.

The Nemesis parts have been provided free of charge for RC hobbyists. The STL files come in a zip archive and can be downloaded from [Thingiverse](#). Included in the archive is a range of accessories for optional items and FPV parts.

SPECIFICATIONS	
<i>Wingspan</i>	120cm
<i>Length</i>	60cm
<i>Printed weight</i>	700g
<i>Flying weight</i>	1100-1400g
<i>Airfoil</i>	PW51
<i>Wing area</i>	22dm ²
<i>Wing loading</i>	50-64g/dm ²

Nemesis Features

- Forward-swept wing
- Thin semi-symmetrical reflexed airfoil
- Twin tractor motors
- Sleek spacious fuselage
- Sliding magnetic canopy system
- Cooling intakes & outlets
- Secure battery attachment plate
- Optional single tail fin fuselage



EQUIPMENT

Carbon fiber

- 6mm x 1000mm tube (1)
- 6mm x 300mm tube (1)
- 3mm x 320mm rod (2)
- 3mm x 100mm rod (2)
- 3mm x 30mm rod (4)

Hardware

- Hex head screw M3 x 10mm (4)
- Hex head screw M3 x 8mm (8)
- Magnet round 6mm x 3mm (2)
- Brass heat-set insert M3 x 5.7mm x 4.6mm (8)
- Battery strap 20mm x 200mm (1)
- Velcro adhesive strip 25mm x 150mm (1)
- Push rods, ball links, clevises (2)

Servos

Two 12-13g mini sized metal geared servos are required. Measuring approx. 24x11.6x24mm.

[EMAX ES08MD II](#) servos are recommended.

Motors

Two 28mm O.D. 2207 or 34mm O.D. 2807 quadcopter style motors are required. 2500kv for 4S builds, 1900kv for 6S builds. Weight ranging from 35-60g. 16x16mm, 19x16mm or 19x19mm bolt patterns.

[T-Motor F60 PRO V 2550kv](#) are recommended for 4S.

[T-Motor F60 PRO V 1950kv](#) are recommended for 6S.

Propellers

5/6/7inch twin or triple blade propellers can be used.

[APC Speed Electric](#) propellers are recommended.

ESCs

Two BLHeli 40-70A ESCs or a BLHeli 4-in-1 ESC can be used.

[SpeedyBee BLS 60A 4-in-1](#) ESC is recommended for moderately fast builds.

[DYS Aria 70A](#) ESCs are recommended for very fast builds.

Battery

It is recommended to use the following 6 cell or 4 cell batteries. Weight ranging from 200-450g:

- 6S 1300-2200 mAh LiPo
- 4S 1500-3000 mAh LiPo
- 4S1P 21700 4000-5000 mAh Li-ion

[CNHL Black Series 1500 mAh 6S 130C](#) LiPo is recommended.

Glue

There are two types of glue used in the construction of the Nemesis.

Elastic adhesive - this type of glue provides a strong bond but can be separated if required. This makes it suitable for attaching parts that can be replaced when damaged. It is recommended to use an elastic adhesive for the following joins:

- Nose to the fuselage
- Battery plate to the fuselage
- Tail fins to the fuselage
- Winglets to the wings
- Wings to the fuselage

[UHU Por](#), [UHU Expanded Polystyrene](#) & [E6000](#) are recommended.

Super glue – this glue is used for all other joins. It is a very strong permanent bond that cannot be easily separated. It is recommended to use a medium formula.

[ZAP-A-GAP Medium CA+](#) is recommended.

PRINTING

Print bed

The Nemesis parts will fit onto a print bed no smaller than 210x210x210mm. This will be suitable for common printers such as the Creality Ender 3/5, Prusa MK3/4 & Bambu Lab range.

Filament

It is recommended to print the Nemesis with PLA+, ABS, ASA or PETG filament. Approximately 600g of filament is required. [ESUN](#) & [Polymaker](#) brands are recommended.

PLA+ is the recommended filament. It is very easy to print and inexpensive compared to other filaments. It is fairly strong & rigid which makes it suitable for fast flying wings.

ABS & ASA filaments offer greater durability, lighter weight and better UV & heat resistance compared to PLA+ however they can be very difficult to print. These filaments are prone to warping especially when printing long thin parts such as wing panels.

PETG is easy to print and provides excellent UV & heat resistance however it is not as rigid as PLA+, ABS or ASA. This filament is best used for the canopy parts only.

Light-weight filaments are not recommended. These filaments are too weak for fast flying models such as the Nemesis. The weight savings of light-weight filaments are not so critical for flying wings with a high power-to-weight ratio.

Tolerances

The Nemesis parts have been drawn with very tight tolerances. This is to ensure that the spars fit securely into the wings & fuselage and there is no slop in the elevons.

Before printing the entire set of parts, it is highly recommended to print a single wing panel & an elevon piece first to check that your carbon fiber tubes & rods fit into the holes. Due to variations between printers, slicers, filament & carbon fiber, it is difficult to design one set of parts to suit everyone so sometimes the holes can be too tight. Please contact Stunt Double if you need help with custom parts.

Slicer settings

Support can be enabled for all parts. Tree style support is recommended. A brim should be used on taller/wider parts to prevent warping and bed detachment. Cubic, adaptive cubic or gyroid infill patterns are recommended.

Settings for the nozzle temperature, bed temperature, flow, speed, retraction & cooling are very important and need to be tuned for your printer and filament.

Pay attention to the Z seam position for each part after slicing. The Z seam can be quite noticeable on single wall prints so it's best rotate the part within the slicer or adjust the settings to hide it as much as possible.

SLICER PROFILES

There are two main slicer profiles recommended for the Nemesis. These profiles have been thoroughly tested to provide the right amount of weight & strength for each part. The two profiles are mostly similar, the main differences being the Wall line count & Infill density settings.

Slicer Profile #1

The first profile is for printing light-weight single wall prints with a small amount of infill. This profile is used for most of the Nemesis parts.

SLICER PROFILE #1	
Layer height	0.2mm
Line width	0.42mm
Wall line count	1
Top layers	3
Bottom layers	3
Infill density	4%
Infill pattern	Cubic
Support	Enabled
Brim	Enabled

Parts to be printed with Slicer Profile #1:

- Canopy Front
- Canopy Rear
- Elevon Inner L & R (15 bottom layers)
- Elevon Outer L & R
- Fuselage Mid
- Fuselage Rear
- Hatch
- Motor Nacelle L & R (35 top layers)
- Nose
- Servo Cover L & R
- Tail Fin
- Wing Inner L & R
- Wing Outer L & R
- Winglet L & R

Slicer Profile #2

The second profile is for printing double wall prints with a high amount of infill. This profile is used for parts that require additional strength. These parts are printed much heavier compared to those printed with the first profile.

SLICER PROFILE #2	
Layer height	0.2mm
Line width	0.42mm
Wall line count	2
Top layers	3
Bottom layers	3
Infill density	50%
Infill pattern	Cubic
Support	Enabled
Brim	Disabled

Parts to be printed with Slicer Profile #2:

- Battery Plate
- Flight Controller Tray
- Fuselage Skid
- Motor Mount
- Servo Tray L & R

Slicer Profile Adjustments

The following parts can benefit from small adjustments to the slicer profiles:

- Elevon Inner L & R - these parts should be printed with 15 bottom layers (15 x 0.2mm). This will create a strong solid elevon control horn.
- Motor Nacelle L & R – these parts should be printed with 35 top layers (35 x 0.2mm). This will create a solid piece of material for the heat-set inserts to be melted in to.
- Tail Fin – these parts can be re-printed with increased Infill density if extra tail weight is needed for balancing.

ASSEMBLY

1. Print the parts

Print the STL parts using the recommended slicer settings.

2. Clean the parts

Remove any leftover support or brim material.

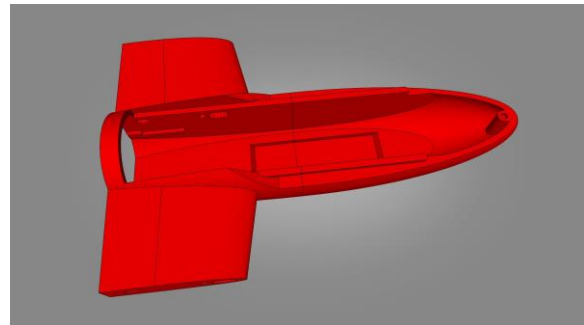
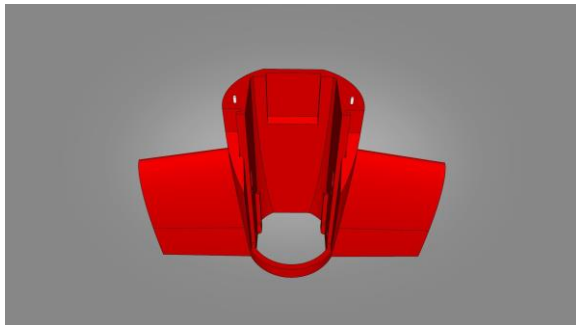
3. Prepare the carbon fiber

Cut the following carbon fiber tubes & rods to length:

- 6mm x 1000mm tube (1)
- 6mm x 300mm tube (1)
- 3mm x 320mm rod (2)
- 3mm x 100mm rod (2)
- 3mm x 30mm rod (4)

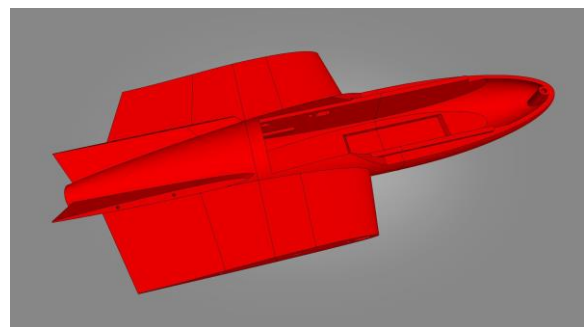
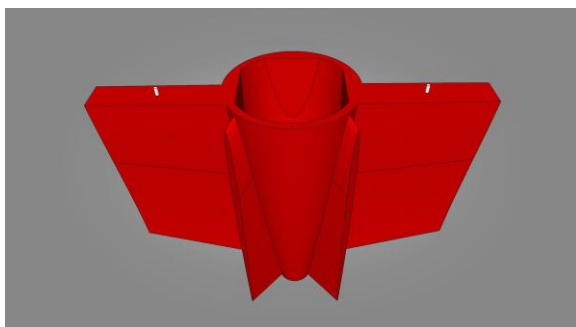
4. Attach the nose

Apply super glue to the alignment holes in the top of the fuselage mid. Insert pieces of filament into the holes and trim to a 5mm length. Apply elastic adhesive to the end of the nose and join it together with the fuselage mid.



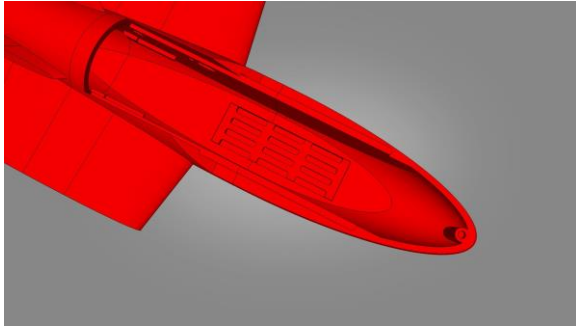
5. Join the fuselage halves

Apply super glue to the alignment holes in the end of the fuselage rear. Insert pieces of filament into the holes and trim to a 5mm length. Apply super glue to the end of the fuselage rear and join it together with the fuselage mid.



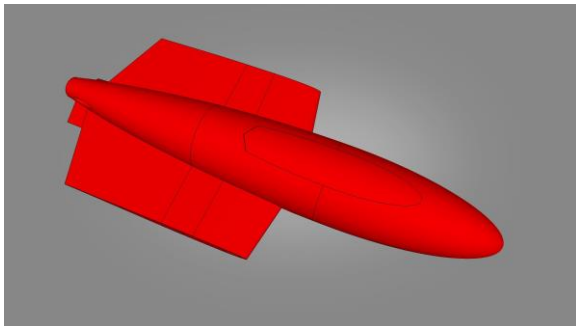
6. Install the battery plate

Apply elastic adhesive to the bottom of the battery plate and insert it into the fuselage.



7. Install the fuselage skid

Apply elastic adhesive to the fuselage skid and attach it to the bottom of the fuselage.



8. Install the battery strap

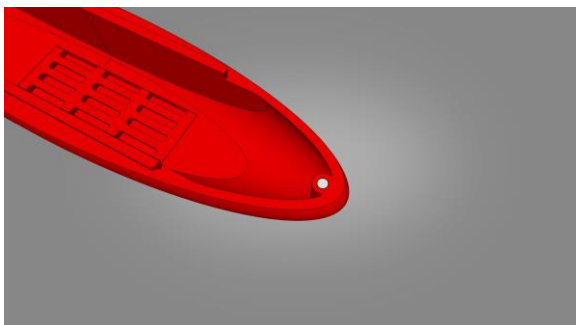
Add a battery strap to the battery plate.

9. Install the battery velcro

Add a 150mm length of velcro adhesive strip to the top of the battery plate.

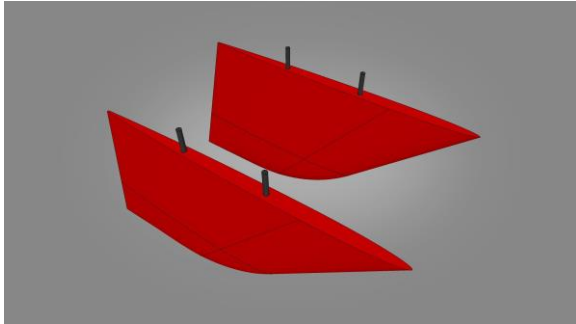
10. Install the nose magnet

Apply super glue to the hole in the front of the nose and insert a 6mm x 3mm round magnet.



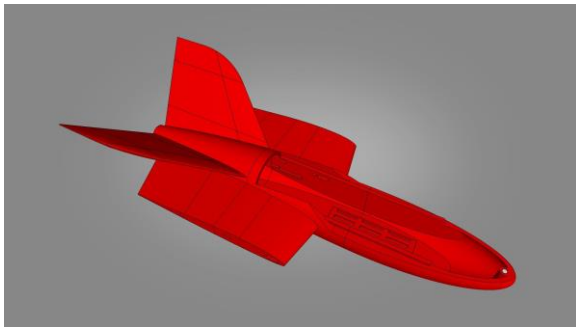
11. Prepare the tail fins

Apply super glue to the holes in the bottom of the tail fins and insert two 3mm x 30mm carbon fiber rods.



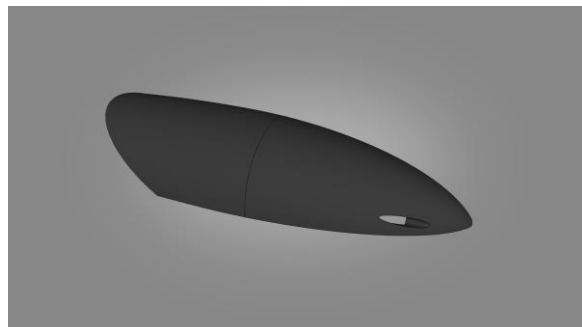
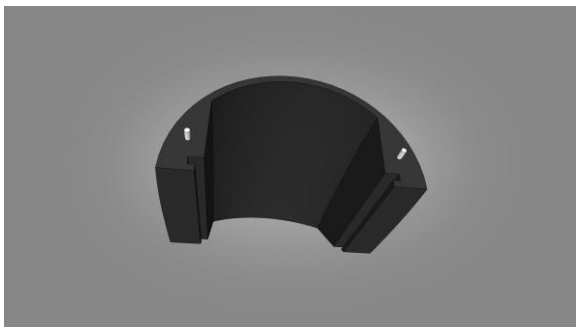
12. Install the tail fins

Apply elastic adhesive to the bottom of the tail fins and attach them to the fuselage.



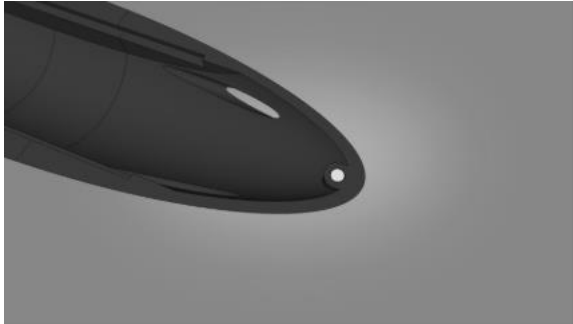
13. Join the canopy

Apply super glue to the alignment holes in the end of the canopy rear. Insert pieces of filament into the holes and trim them to a 5mm length. Apply super glue to the end of the canopy rear and join it together with the canopy front.



14. Install the canopy magnet

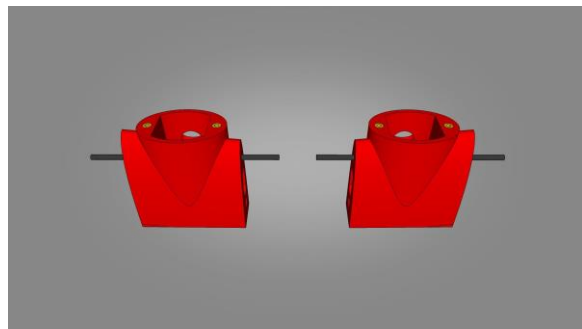
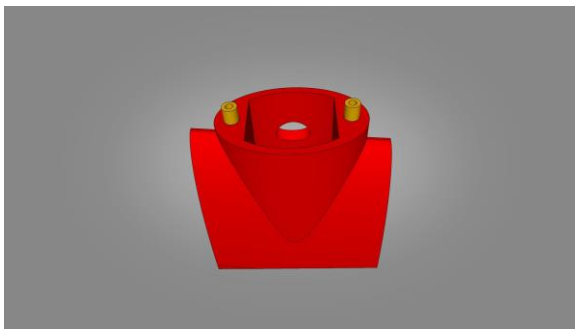
Apply super glue to the hole underneath the canopy and insert a 6mm x 3mm round magnet.



Check the polarity of the magnet against the fuselage magnet before installing.

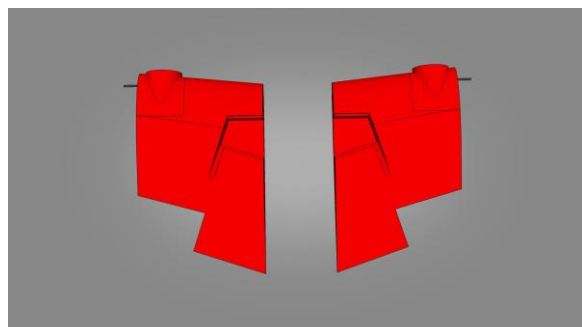
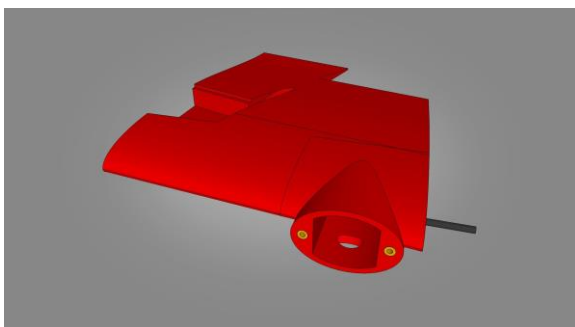
15. Prepare the motor nacelles

Insert two M3 x 5.7mm x 4.6mm brass heat-set inserts into the holes on top of the motor nacelles. Press the inserts into position using a soldering iron. Insert 3mm x 100mm carbon fiber rods into the sides of the motor nacelles. Add a few drops of super glue to secure the rods.



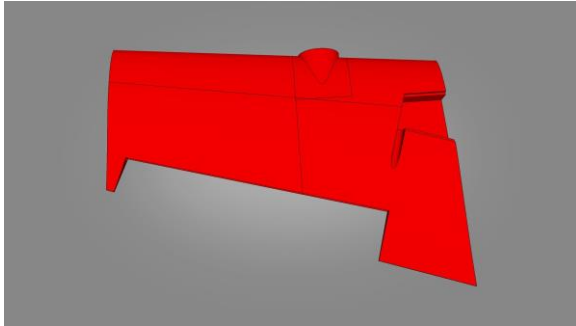
16. Attach the motor nacelles

Apply super glue to the motor nacelles and attach them to the wing inners.



17. Join the wings

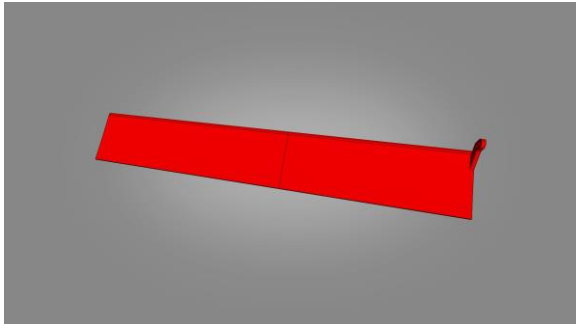
Apply super glue to the ends of the wing inners and join them with the wing outers.



A 6mm carbon fiber tube can be inserted into the wings to help with alignment. Be careful not to glue the tube inside the wing.

18. Prepare the elevons

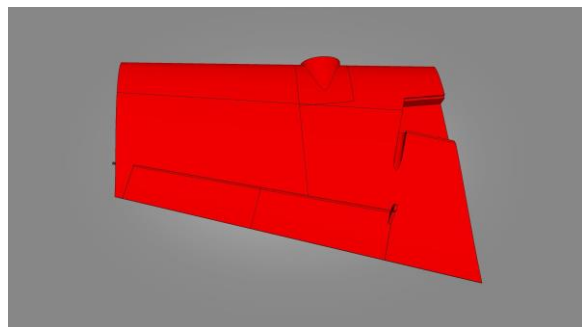
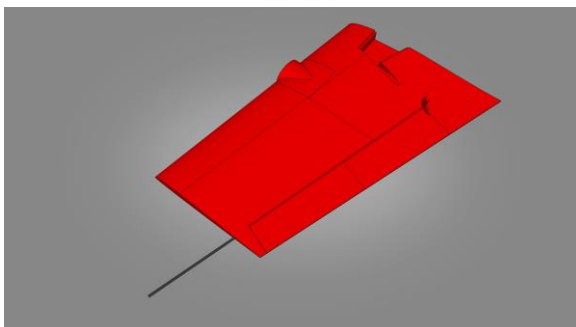
Apply super glue to the top of the elevon inners and join them with the elevon outers.



A 3mm carbon fiber rod can be inserted into the elevon to help with alignment. Be careful not to glue the rod inside the elevon.

19. Attach the elevons

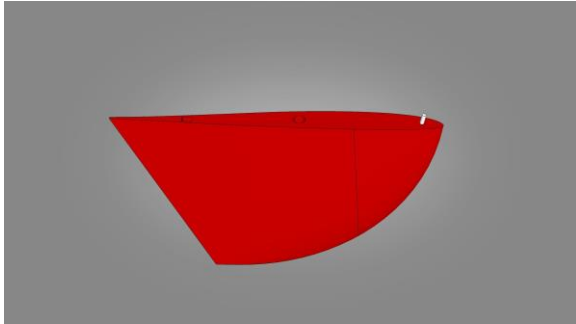
Align the elevons with the wing panels. Insert 3mm x 320mm carbon fiber rods into the holes in the wing tips and push them through the elevons into the wing root.



Check that the elevons move freely. The edges of the elevons can be lightly sanded if needed.

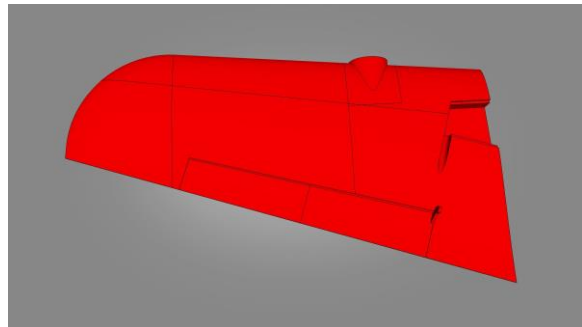
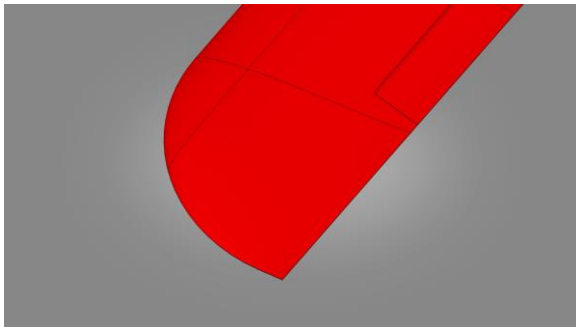
20. Prepare the winglets

Apply super glue to the alignment holes in the winglets. Insert pieces of filament into the holes and trim to a 5mm length.



21. Attach the winglets

Apply elastic adhesive to the end of the winglets and attach them to the wings.



22. Attach the motors

Attach the motors to the motor mounts using four M3 x 8mm hex head screws.

23. Install the motors

Feed the motor wires through the holes in the bottom of the motor nacelles then through the channels in the wings. Attach the motor mounts to the motor nacelles using two M3 x 10mm hex head screws.

Extend the motor wires if needed.

24. Prepare the servos

Center the servos and attach a control arm at a 90-degree angle. Attach a clevis to the servo control arm. Apply elastic adhesive to the bottom of the servos and attach them to the servo trays.

25. Install the servos

Apply elastic adhesive to the bottom of the servo trays and insert them into the wings. Feed the servo wires through the holes in the end of the wings.

Allow the glue to fully set before connecting the linkages.

26. Install the linkages

Attach a ball link to a threaded rod. Connect the threaded rod to the servo clevis. Attach the ball link to the elevon control horn and set 2mm of up reflex.

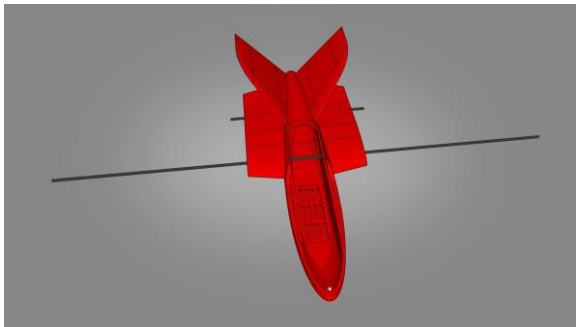
Adding reflex is important.

27. Install the servo covers

Apply elastic adhesive to the servo covers and attach them to the wings.

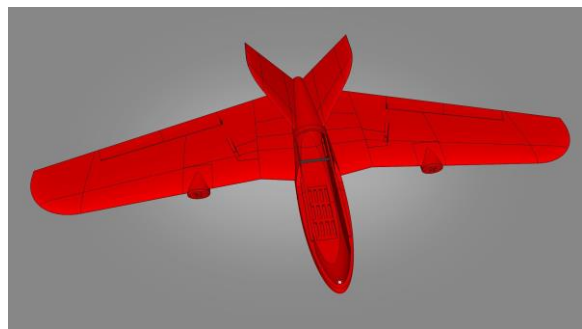
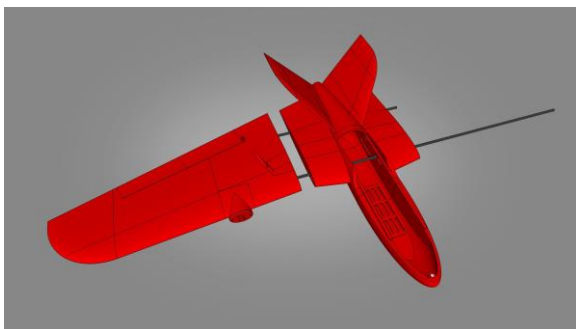
28. Attach the wing spars

Insert the 6mm x 1000mm carbon fiber tube into the front of the fuselage and the 6mm x 300mm tube into the rear.



29. Attach the wings

Apply elastic adhesive to the sides of the fuselage. Connect the wings with the wing spars and join them together with the fuselage.

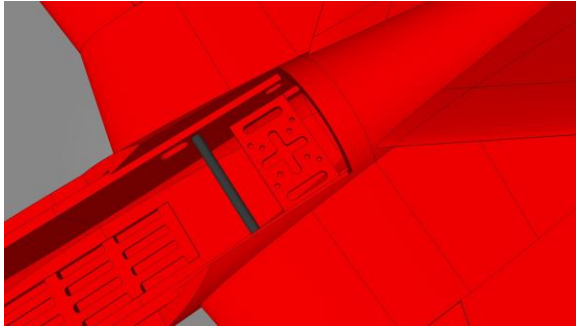


30. Install the ESCs

Solder the motor wires to the ESCs. Secure the ESCs inside the fuselage using double-sided adhesive tape.

31. Install the flight controller tray

Apply super glue to the bottom of the flight controller tray and insert it into the fuselage.



If needed, add M3 spacers to the tray before installing it into the fuselage.

32. Install the electronic gear

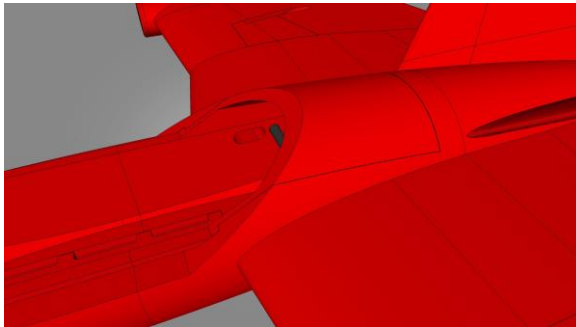
Install any electronic gear such as a receiver, BEC, flight controller, GPS, FPV gear, etc.

33. Attach the propellers

Attach CW & CCW propellers to the motors.

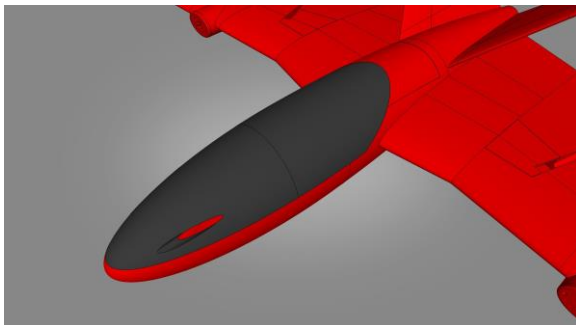
34. Attach the hatch

Slide the hatch onto the fuselage rails.



35. Attach the canopy

Slide the canopy onto the fuselage rails.



Make sure the canopy super glue has fully set before attempting to slide the canopy on to the fuselage for the first time.

SETUP

Elevon throw

Roll = +/- 12mm

Pitch = +/- 6mm

Elevon reflex

2mm up

Adding reflex to the elevons is very important. The model is vulnerable to crashing on the maiden flight if no reflex is added.

Center of gravity

18mm behind the leading-edge join of the wing and fuselage.

There are CG notches positioned underneath the wing to help set the balance. The model can be safely flown +/- 5mm from the marked CG position.

ACCESSORIES

The following optional parts are available in the "Accessories" folder:

Alternate parts

- Fuselage Rear Twin Tail Fins
- Fuselage Skid Flat
- Motor Mount 16x16mm
- Motor Mount 19x19mm
- Wing Outer Led Strip L
- Wing Outer Led Strip R

FPV canopy parts

- Canopy Front AtomRC Gimbal
- Canopy Front DJI 03 Air
- Canopy Front FPV Camera 14x14mm
- Canopy Front FPV Camera 19x19mm
- Canopy Front Pan Servo Backward
- Canopy Front Pan Servo Forward
- Canopy Front RunCam Link 19mm Camera
- Canopy Rear AtomRC Gimbal
- Canopy Rear DJI 03 Air

Walksnail canopy parts

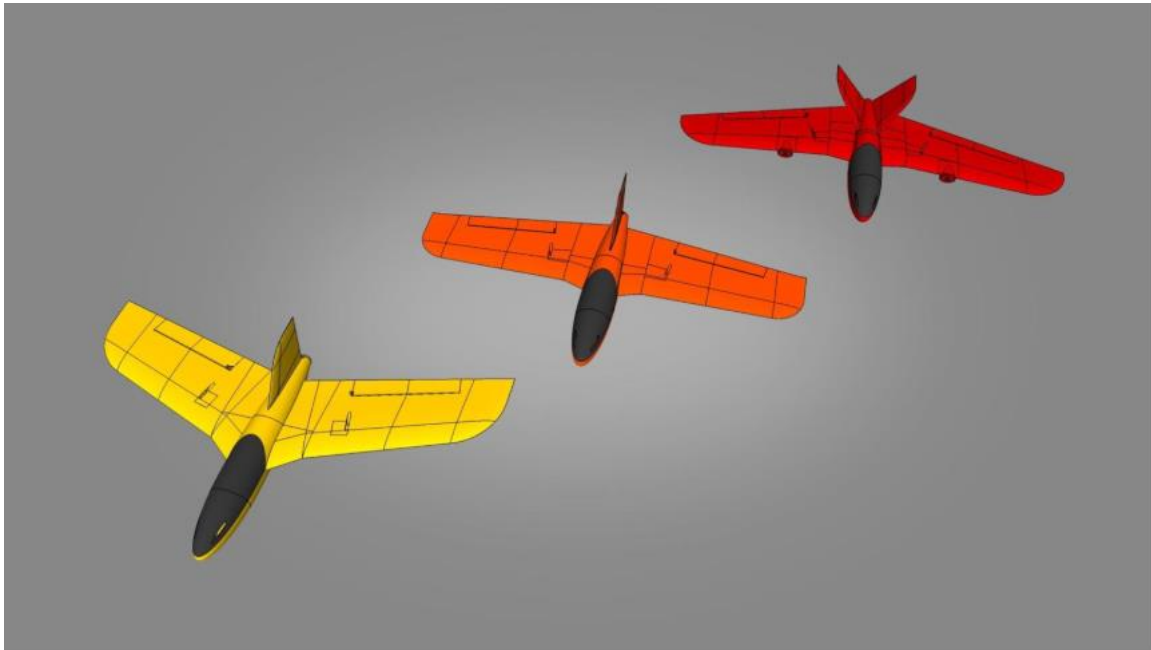
- Canopy Front Walksnail Camera 14x14mm Rearwards
- Canopy Front Walksnail Camera 14x14mm
- Canopy Front Walksnail Camera 19x19mm Rearwards
- Canopy Front Walksnail Camera 19x19mm
- Canopy Rear Walksnail VTX V1
- Canopy Rear Walksnail VTX V2

Custom accessories are available for a small fee. Please contact Stunt Double for more info.

COMPATIBILITY

The Nemesis shares the same canopy as Stunt Double's [Firebird V2](#) & [Stormbird](#) models. This means that the same canopy can be used for all three planes. For example, a single FPV canopy can be built up then easily swapped around.

The three planes share an identical fuselage but have their own wing shape which offers different flight characteristics. Having a single canopy is a cost-effective way to have three different FPV planes ready to go at the same time.



	AGILITY	EFFICIENCY	STABILITY	SPEED
<i>Firebird V2</i>	★★★★★	★★★★★	★★★★★	★★★★
<i>Stormbird</i>	★★★★★	★★★★★★	★★★★	★★★★★
<i>Nemesis</i>	★★★★	★★★★★	★★★★★★	★★★★

SUPPORT

Facebook

www.facebook.com/groups/3dprintedrcplanes

The Facebook group “3D Printed RC Planes by Stunt Double” is the main support area for Stunt Double’s models. It is an active group with over 11,000 members. This group can be used for help, sharing builds pics, uploading videos and requesting custom accessories.

RCGroups

www.rcgroups.com/forums/showthread.php?4498705-Flying-Wings-by-Stunt-Double

The RCGroups thread “Flying Wings by Stunt Double” is useful for those who do not use Facebook.

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